AI and Generative AI Fundamentals - Study Notes

Original study notes prepared for the DataMind RAG Assistant project.

What is artificial intelligence?

Artificial intelligence (AI) broadly refers to computer systems that perform tasks which typically require human intelligence, such as understanding language, recognizing images, or making decisions under uncertainty. Machine learning is the dominant modern approach to building AI systems, using data-driven pattern learning rather than hand-coded rules.

Deep learning is a subfield of machine learning built on neural networks with many layers, which has driven much of the recent progress in computer vision, speech recognition, and natural language processing.

What is generative AI?

Generative AI refers to models that can produce new content — text, images, audio, or code — rather than simply classifying or predicting a single number or label. Large language models (LLMs) are a prominent example: they are trained on huge amounts of text to predict the next word in a sequence, which lets them generate coherent, contextually relevant text.

Other generative model families include diffusion models, widely used for image generation, and generative adversarial networks (GANs), an earlier architecture where two networks — a generator and a discriminator — are trained together in competition.

Retrieval-augmented generation (RAG)

Retrieval-augmented generation combines a language model with an external knowledge source: instead of relying purely on what the model memorized during training, the system first retrieves relevant documents or passages from a knowledge base, then asks the model to answer using that retrieved context.

This approach helps ground answers in verifiable, up-to-date source material, reduces the chance of the model inventing plausible-sounding but incorrect facts (a failure mode often called 'hallucination'), and makes it possible to cite the specific source and page an answer came from — which is exactly the architecture this project (DataMind) implements.

Embeddings and vector search

An embedding is a numeric vector representation of a piece of text (or image, or other data) such that semantically similar inputs end up with similar vectors. Embedding models are typically trained so that the distance between two vectors reflects how related their meanings are, not just whether they share exact words.

A vector database stores these embeddings and lets you efficiently find the vectors most similar to a given query vector — this is the 'retrieval' half of retrieval-augmented generation, and is what allows a system to find the most relevant document chunks for a user's question out of a large corpus.